

EXERCISES AND EXACT SOLUTIONS

Outbreak Signals Before Alarm

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C21
Notebook	V7_B_N01
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Add a moving-baseline method.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Test threshold sensitivity.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Define a confirmation workflow.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 4

Explain why public communication cannot be automated from this notebook.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Use only prior weeks, preserve seasonality, and avoid contaminating baselines with active events.

Solution 2

Compare verified-event sensitivity, false alerts, delay, workload, and subgroup/geographic effects.

Solution 3

Assign triage, data verification, epidemiological assessment, laboratory/clinical review, authorization, communication, closure, and retrospective review.

Solution 4

Communication carries legal, scientific, behavioural, privacy, and political consequences and requires authorized contextual judgement.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated